THE TWO-TERM CECH SEQUENCE OF A MODULE FOR A COVERING BY BASIC OPENS IS EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- Exactness can be checked after localising at maximal ideals. Invoked at the start of the counted proof (lines 4540-4542). Reduces the exactness of the displayed sequence to the local case.
- Localising a localisation by a product of the covering elements. Used throughout the local computation (lines 4543-4567). Simplifies the terms of the sequence once one of the covering elements is invertible.

Source reference. Original lemma 00HN: sources/stacks-algebra/pinned-original-source.tex, source lines 4268–4268. Navigation pointer for \ref{lemma-characterize-zero-local} used inside the counted span (line 4542); the argument that consumes it is retained in the span.

Source reference. Original lemma 07K0: sources/stacks-algebra/pinned-original-source.tex, source lines 1388–1388. Navigation pointer for \ref{lemma-universal-property-localization-module} used inside the counted span (line 4554); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02C7: sources/stacks-algebra/pinned-original-source.tex, source lines 1519–1519. Navigation pointer for \ref{proposition-localize-twice-module} used inside the counted span (line 4548); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $f_1, \ldots, f_n$ be elements of R generating the unit ideal. Let M be an R-module. The sequence*

$$0 \to M \xrightarrow{\alpha} \bigoplus\nolimits_{i=1}^{n} M_{f_i} \xrightarrow{\beta} \bigoplus\nolimits_{i,j=1}^{n} M_{f_i f_j}$$

is exact, where $\alpha(m) = (m/1, \ldots, m/1)$ and $\beta(m_1/f_1^{e_1}, \ldots, m_n/f_n^{e_n}) = (m_i/f_i^{e_i} - m_j/f_j^{e_j})_{(i,j)}$.

Proof. It suffices to show that the localization of the sequence at any maximal ideal $\mathfrak{m}$ is exact, see Lemma 00HN. Since $f_1, \ldots, f_n$ generate the unit ideal, there is an i such that $f_i \notin \mathfrak{m}$. After renumbering we may assume $i = 1$. Note that $(M_{f_i})_\mathfrak{m} = (M_\mathfrak{m})_{f_i}$ and $(M_{f_i f_j})_\mathfrak{m} = (M_\mathfrak{m})_{f_i f_j}$, see Proposition 02C7. In particular $(M_{f_1})_\mathfrak{m} = M_\mathfrak{m}$ and $(M_{f_1 f_i})_\mathfrak{m} = (M_\mathfrak{m})_{f_i}$, because f_1 is a unit. Note that the maps in the sequence are the canonical ones coming from Lemma 07K0 and the identity map on M. Having said all of this, after replacing R by $R_\mathfrak{m}$, M by $M_\mathfrak{m}$, and f_i by their image in $R_\mathfrak{m}$, and f_1 by $1 \in R_\mathfrak{m}$, we reduce to the case where $f_1 = 1$.

Assume $f_1 = 1$. Injectivity of α is now trivial. Let $m = (m_i) \in \bigoplus_{i=1}^n M_{f_i}$ be in the kernel of β. Then $m_1 \in M_{f_1} = M$. Moreover, $\beta(m) = 0$ implies that m_1 and m_i map to the same element of $M_{f_1 f_i} = M_{f_i}$. Thus $\alpha(m_1) = m$ and the proof is complete. □